

PHY 101 – GENERAL PHYSICS I

COMPLETE TEST SUMMARY WITH EXAMPLES AND FORMULAS

This document is a complete and carefully structured summary of PHY 101 for test preparation. All major topics are explained in clear academic language with examples, types, and applications. A full page containing one hundred essential physics formulas is included for memorization and quick revision.

1. Measurement in Physics

Physics is based on measurement, which involves comparing an unknown quantity with a standard unit. Physical quantities are divided into base quantities such as length, mass, and time, and derived quantities such as velocity, acceleration, and force. The SI system is the internationally accepted system of units. Accuracy refers to closeness to the true value, while precision refers to repeatability. Errors in measurement may be systematic, random, or gross. Example: measuring length with a meter rule introduces possible zero error.

2. Scalars and Vectors

Scalar quantities have magnitude only, such as mass, temperature, and time. Vector quantities have both magnitude and direction, such as displacement, velocity, and force. Vectors may be added using graphical or analytical methods. Resolution of vectors involves breaking a vector into perpendicular components. Example: resolving a force acting at an angle into horizontal and vertical components.

3. Kinematics

Kinematics is the study of motion without considering its cause. It includes displacement, velocity, acceleration, and equations of motion. Motion may occur in one dimension, two dimensions, or three dimensions. Example: a car accelerating uniformly along a straight road obeys the equations of linear motion.

4. Dynamics

Dynamics deals with the causes of motion. Newton's First Law explains inertia, the Second Law relates force to mass and acceleration, and the Third Law explains action and reaction. Forces include gravitational force, frictional force, tension, and normal reaction. Example: friction opposes motion between two surfaces in contact.

5. Work, Energy, and Power

Work is done when a force causes displacement. Energy is the capacity to do work and exists as kinetic and potential energy. The law of conservation of energy states that energy cannot be created or destroyed. Power is the rate of doing work. Example: a falling object converts potential energy to kinetic energy.

6. Momentum and Collisions

Momentum is the product of mass and velocity. The principle of conservation of momentum states that total momentum of a closed system remains constant. Collisions may be elastic or inelastic. Example: recoil of a gun illustrates momentum conservation.

7. Circular Motion

Circular motion occurs when an object moves in a circular path. It requires a centripetal force directed towards the center of the circle. Example: motion of satellites around the Earth.

8. Rotational Motion

Rotational motion involves angular displacement, angular velocity, and angular acceleration. Torque causes rotation, and moment of inertia depends on mass distribution. Example: rotation of a wheel about its axis.

PHY 101 – ONE HUNDRED ESSENTIAL FORMULAS

1. $v = u + at$
2. $s = ut + \frac{1}{2} at^2$
3. $v^2 = u^2 + 2as$
4. $a = (v - u)/t$
5. $s = (u + v)t/2$
6. $F = ma$
7. $W = mg$
8. $f = \mu R$
9. $P = F/A$
10. $F = Gm_1m_2/r^2$
11. $W = Fs$
12. $K.E = \frac{1}{2} mv^2$
13. $P.E = mgh$
14. $P = W/t$
15. Efficiency = output/input
16. $p = mv$
17. Impulse = Ft
18. $F = \Delta p/t$
19. $v = s/t$
20. $\rho = m/V$
21. $\omega = \theta/t$
22. $v = r\omega$
23. $a = v^2/r$
24. $T = 2\pi/\omega$
25. $F = mv^2/r$
26. $\tau = Fr$
27. $I = mr^2$
28. $K(\text{rot}) = \frac{1}{2} I\omega^2$
29. $P = \tau\omega$

30. $L = I\omega$
31. $R = V/I$
32. $V = IR$
33. $P = VI$
34. $E = VIt$
35. $Q = It$
36. $\lambda = h/p$
37. $E = hf$
38. $c = f\lambda$
39. $n = c/v$
40. Magnification = image/object
41. Pressure = F/A
42. Boyle: $P_1V_1 = P_2V_2$
43. Charles: $V/T = \text{constant}$
44. Heat = $mc\Delta\theta$
45. Latent heat = mL
46. Stress = F/A
47. Strain = extension/original length
48. Young modulus = stress/strain
49. Frequency = $1/T$
50. Angular momentum = mvr
51. Work = change in energy
52. Force = rate of change of momentum
53. Average speed = total distance/total time
54. Acceleration = change in velocity/time
55. Centripetal acceleration = v^2/r
56. Escape velocity = $\sqrt{(2GM/R)}$
57. Gravitational potential = $-GM/r$
58. Moment = force $\times$ distance
59. Mechanical advantage = load/effort

60. Velocity ratio = distance moved effort/load

61. Efficiency = MA/VR

62. Wave speed = $f\lambda$

63. Intensity = power/area

64. Electric field = F/q

65. Potential difference = W/q

66. Capacitance = Q/V

67. Energy stored in capacitor = $\frac{1}{2} CV^2$

68. Power loss = I^2R

69. Resistivity = RA/L

70. Angular acceleration = $(\omega - \omega_{\text{initial}})/t$

71. Linear momentum = mv

72. Hooke's law: $F = kx$